

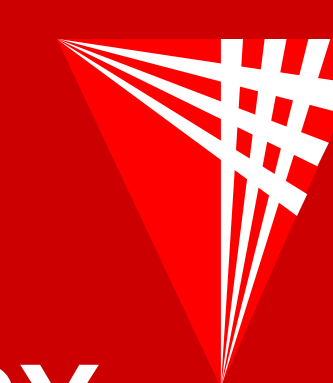
# Multi-Node Inertial Motion Capture Device

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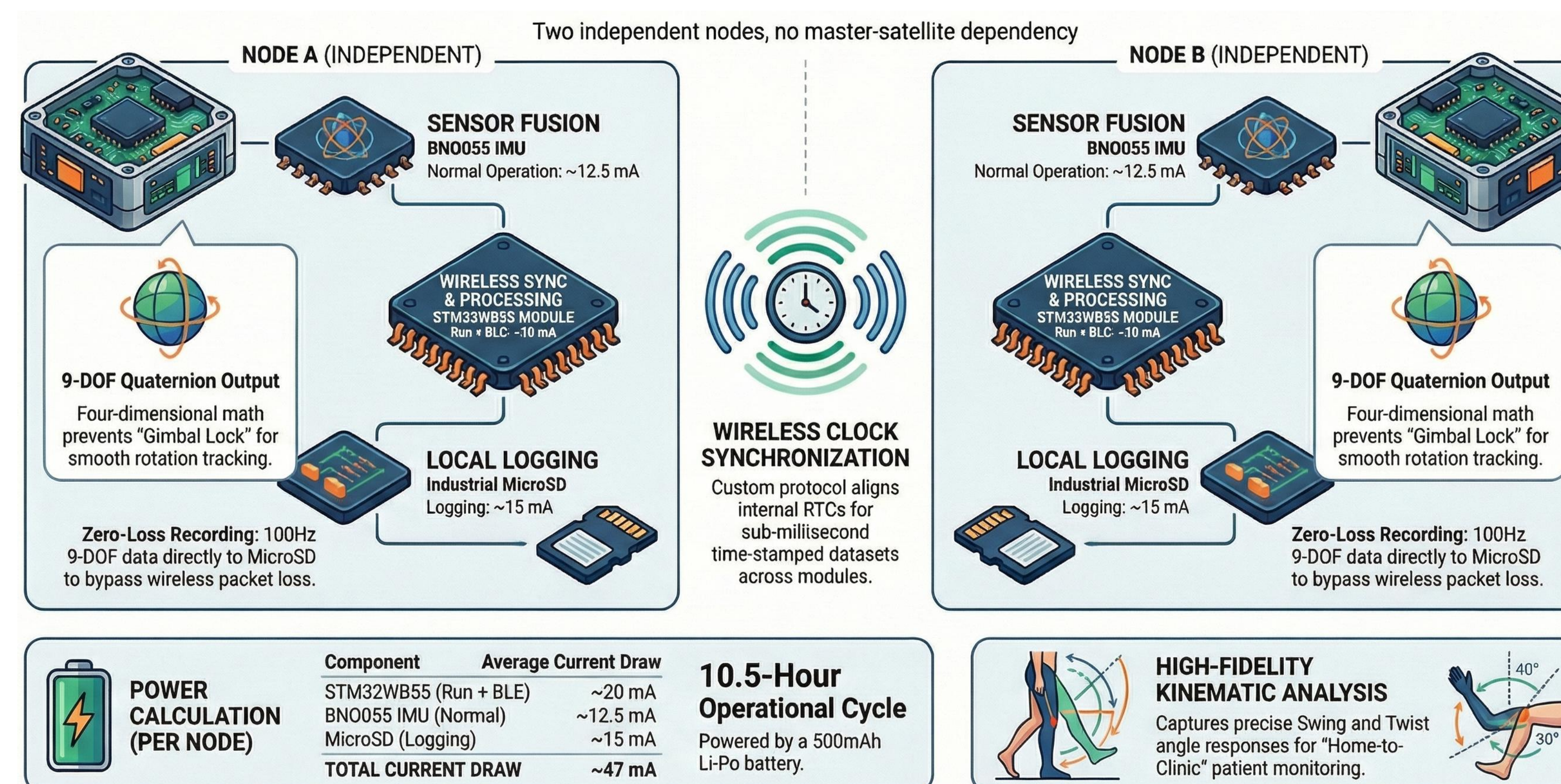
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## INTRODUCTION

The global rise in neurological disorders, such as epilepsy and post-stroke hemiparesis, creates a critical need for continuous, objective motion tracking to accurately assess recovery and detect life-critical events. This research introduces a decentralized, dual-node system utilizing the STM32WB55 microcontroller for synchronized 9-DOF kinematic tracking to meet this clinical demand. By replacing centralized architectures with a custom wireless clock-sync protocol, the system achieves medical-grade reliability and autonomous operation without the risk of packet loss common in Bluetooth-dependent devices. The architecture leverages high-speed SPI edge-logging to industrial-grade MicroSD cards, ensuring total data retention of tri-axial acceleration and orientation quaternions. Optimized for clinical utility, the standalone device provides a 10-hour runtime on a 500mAh Li-Po source. This platform enables real-world monitoring of patient recovery and abnormal movement patterns outside of traditional laboratory settings.

## BACKGROUND AND APPROACH



## CURRENT WORK

- **Decentralized Node Architecture:** Transitioned from a centralized master-slave model to an independent node architecture using STM32WB55 microcontrollers. This eliminates dependencies and prevents the wireless bottleneck.
- **High-Resolution Data Logging:** Achieved a deterministic 100Hz sampling rate, logging 9-DOF data (tri-axial acceleration, angular velocity, and quaternions) directly to industrial-grade MicroSD cards via high-speed SPI.
- **Power Management:** Validated an operational window of approximately 10 hours using 500mAh Li-Po batteries, meeting the requirement for a full clinical monitoring day.
- **Physical Prototype Fabrication:** Manufactured a functional Power Distribution Board (PDB) incorporating a TP4056 charging circuit. The current prototype is 150mm x 150mm.
- **Robotic Validation Rig:** Constructed a 2-DOF test rig using SG92R servo motors and an Arduino Uno to provide a "Mechanical Ground Truth" for verifying IMU accuracy.
- **Software & Analysis Pipeline:** Developed a firmware stack in Arduino IDE and a post-processing analysis pipeline in VS Code (Python) using Pandas and Scipy to cross-correlate IMU data with the robotic rig's movements.

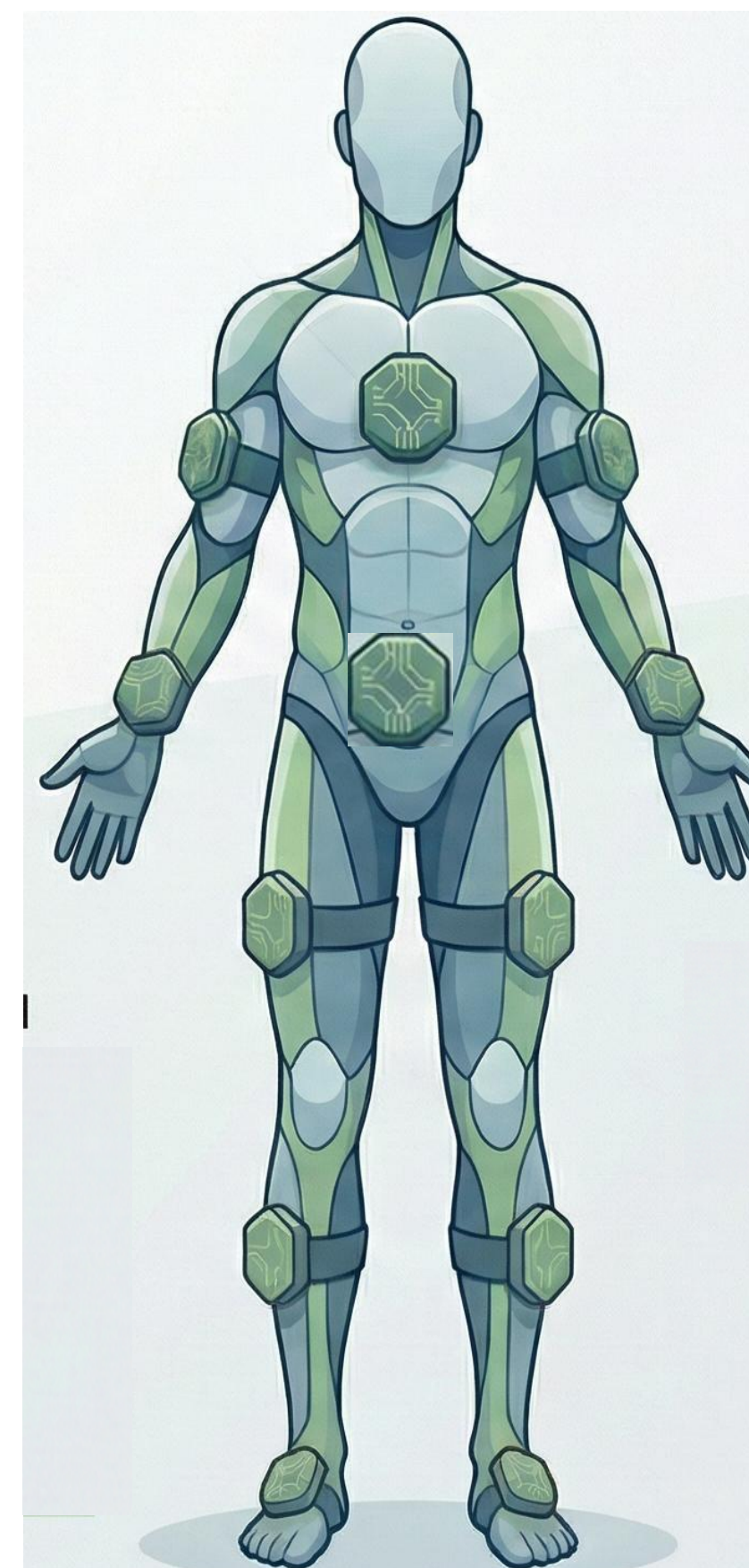
## PROBLEM STATEMENT

- **Data Integrity:** Current technologies fail to provide the high-fidelity, continuous data for managing post-stroke and epilepsy conditions, often due to significant data loss from wireless packet drops during dynamic movement.
- **Architecture & Synchronization:** Traditional centralized architectures create single points of failure, while offsets between independent sensors compromise the accuracy of coordination and gait symmetry analysis.
- **Biomechanical Interference:** Bulky, wired systems impede natural motion and distorted data that does not accurately reflect the patient's biomechanical state.
- **Hardware & Processing:** Low-cost MCUs frequently lack the dual-core power required to manage high-frequency sensor fusion and telemetry, resulting in performance-degrading latency.

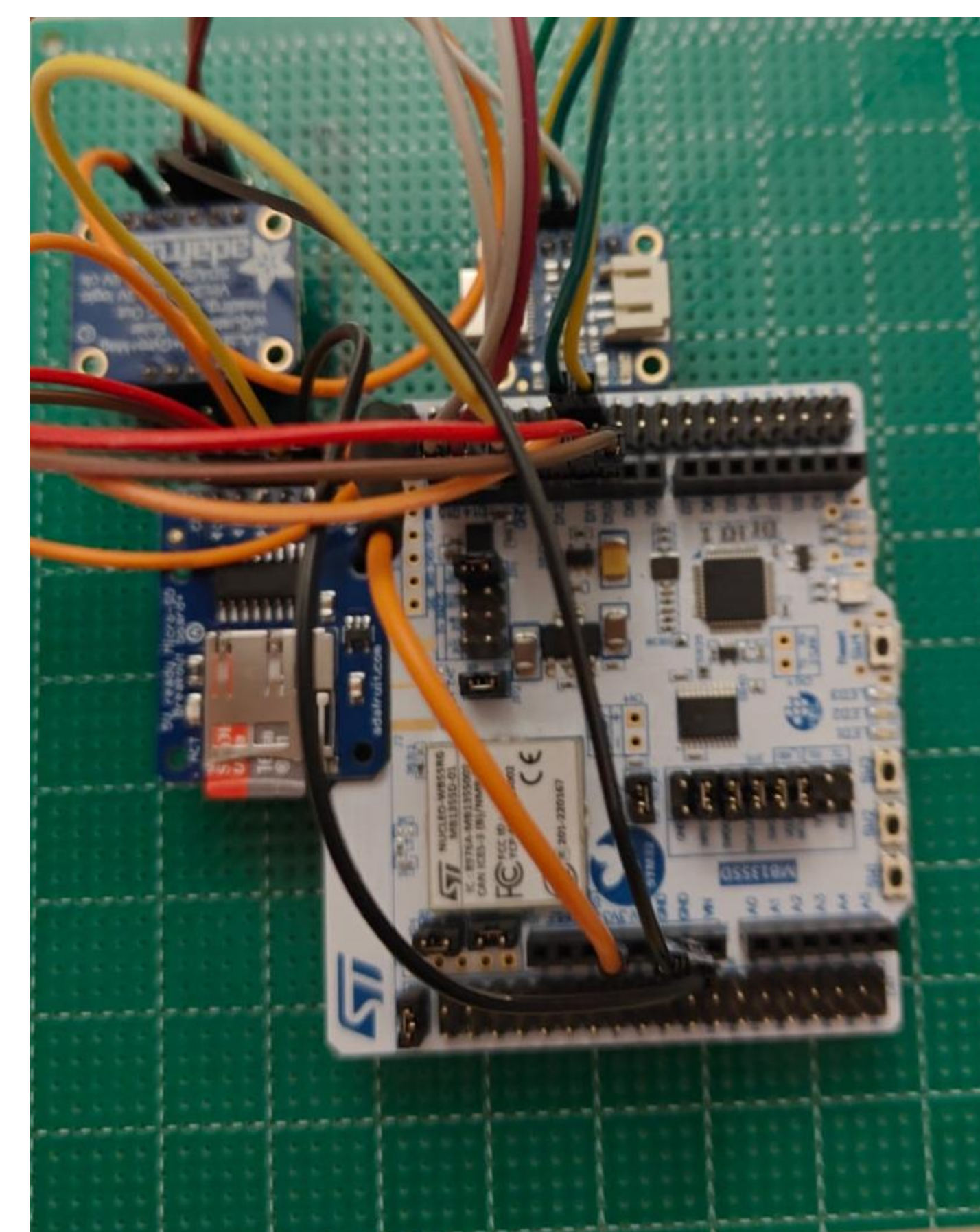
Robotic Validation Rig



Possible Tracking Positions



Power Distribution Board



## FUTURE WORK

- **PCB Miniaturization:** Transitioning from the current prototype to a custom compact PCB. This will reduce the footprint to the target 100mm x 100mm x 50mm, making it truly wearable and unobtrusive for patients.
- **Enhanced Synchronization Protocols:** Implementing sub-GHz hardware synchronization or specialized radio handshakes. This will further reduce micro-jitter between nodes during high-velocity movements.
- **Advanced Magnetometer Calibration:** Robust hard-iron and soft-iron calibration algorithms. This is essential for maintaining heading accuracy in varied magnetic environments, such as homes or clinics with metal equipment.